

M624 HOMEWORK – SPRING 2026

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SET 1 - DUE TUESDAY 02/17/2026 BY 5PM

From Chapter 4 :

Additional Problem 1: Prove the Auxiliary Lemma we stated in Lecture 2. Namely prove the following:

A normed vector space $(X, \|\cdot\|)$ is *complete* if and only if every absolutely convergent series in X converges in X .

Recall; i) A series $\{x_n\}_n$ is said to be absolutely convergent if $\sum_{n=1}^{\infty} \|x_n\| < \infty$.

ii) A series $\sum_{n=1}^{\infty} x_n$ converges in X if there exists an x in X such that $\lim_{N \rightarrow \infty} \sum_{n=1}^N x_n = x$ in X ; that is if $\|\sum_{n=1}^N x_n - x\| \rightarrow 0$ as $N \rightarrow \infty$

Additional Problem 2: For any $1 \leq p < \infty$ consider the space

$$L^p(\mathbb{R}^d) := \{f : \mathbb{R}^d \rightarrow \mathbb{C}, \text{ measurable, } \|f\|_{L^p(\mathbb{R}^d)} := \left(\int_{\mathbb{R}^d} |f(x)|^p dm \right)^{\frac{1}{p}} < \infty\}.$$

Assume that $\|f\|_{L^p(\mathbb{R}^d)}$ is a norm[†] whence $d_p(f, g) := \|f - g\|_{L^p(\mathbb{R}^d)}$ defines a metric and L^p is a metric space. Use the proof of completeness that I gave in class from Folland (Lecture 3) when $p = 2$ to **prove** that $L^p(\mathbb{R}^d)$ is *complete*.

(†) Question: can you guess what you would need to prove the triangle inequality when $p \neq 2$?. Justify your answer.

From Chapter 4 (pp 193-194): 1, 2*, 3, 4 (show completeness only), 5, 6, 7.

* to show that $f - g$ is *orthogonal* to g you need to show that $\langle f - g, g \rangle = 0$.

SET 2 - DUE THURSDAY 02/26/2026 BY 5PM

From Chapter 4 (pp 195-197): 8a)

Pb. I. (This is an undergrad. problem but a very useful property to remember)
Let $\phi : \mathbb{R} \rightarrow \mathbb{R}$ be a periodic function with period p ; that is $\phi(x + p) = \phi(x)$, $\forall x \in \mathbb{R}$. Assume that ϕ is integrable on any finite interval.

(a) Prove that for any $a, b \in \mathbb{R}$

$$\int_a^b \phi(x) dx = \int_{a+p}^{b+p} \phi(x) dx = \int_{a-p}^{b-p} \phi(x) dx$$

(b) Prove that for any $a \in \mathbb{R}$

$$\int_{-p/2}^{p/2} \phi(x+a) dx = \int_{-p/2}^{p/2} \phi(x) dx = \int_{-p/2+a}^{p/2+a} \phi(x) dx$$

In particular we have that $\int_a^{a+p} \phi(x) dx$ does not depend on a , as we discussed in class.

Pb.II. Consider $f \in L^2([-\pi, \pi])$ and assume that $\sum_{n \in \mathbb{Z}} a_n e^{inx} = f(x)$ a.e. x . Show that on any subinterval $[a, b] \subset [-\pi, \pi]$,

$$\int_a^b f(x) dx = \sum_n \int_a^b a_n e^{inx} dx.$$

In particular if $g(x) = \int_a^x f(y) dy$, the Fourier coefficients and series of $g(x)$ can be obtained from a_n , the Fourier coefficients of f .

Pb. III. For $0 < \alpha < 1$, we say that a function f is C^α -Hölder continuous with exponent α if there exists a constant $c = c_\alpha > 0$ such that $|f(x) - f(y)| \leq c|x - y|^\alpha$ for all x, y . For $k \in \mathbb{N}$, we can also define the space $C^{k,\alpha}$ to be that of functions which are k -th times differentiable and whose k -th derivative is C^α -Hölder continuous (we could relabel C^α as $C^{0,\alpha}$). Consider now f a 2π -periodic $C^{k,\alpha}$ function. If a_n are the Fourier coefficients of f , show that for some $C > 0$ independent of n ,

$$|a_n| \leq \frac{C}{|n|^{k+\alpha}}$$

Hint Start by integrating by parts (and using periodicity).

Bonus Problem (do but do not turn in): 2*a)b) from Chapter 4, pp 202.

SET 3 - DUE THURSDAY 03/12/2026 BY 5PM

From Chapter 4 (pp 195-197): 10, 11, 12, 13

Pb. I. Consider the subspace $\mathcal{S}$ of $L^2([0, 1])$ spanned by the functions: 1, x , and x^3 .

a) Find an orthonormal basis of $\mathcal{S}$.

b) Let $P_{\mathcal{S}}$ denote the orthogonal projection on the subspace $\mathcal{S}$, compute $P_{\mathcal{S}}x^2$.

Assigned Reading from Chapter 4 of [Stein-Shakarchi Vol 3]:

a) Examples 1) and 2) on pages 178–180.

b) Please read ahead sections 4.5.1, 4.5.2 and 4.5.3 of section 5 chapter 4.

From Chapter 4 (pp 195-197): 18, 19, 20

SET 4 - DUE THURSDAY 03/26/2026 BY 5PM

Assigned Reading from Chapter 4 of [Stein-Shakarchi Vol 3]:

Remarks (a), (b), (c) on pages 184-185.

From Chapter 4 (pp 187): Read and rewrite filling in all details the Proof of Proposition 5.5 (do but do not turn in)

From Chapter 4 (pp 189): Read and rewrite filling in all details the Proof of Proposition 6.1 (do but do not turn in).

From Chapter 4 (pp 197-202): 21, 22, 23, 25, 26.

SET 5 - DUE THURSDAY 04/02/2026 BY 5PM

From Chapter 4 (pp 197-202): 28 (for c) consider only the *if* part), 30, 32, 33.

Do but do not turn in. From Chapter 4: 29 (p 199-200) and 6 in Section 8 (p. 203-204). The Fredholm's Alternative for compact operators holds generally on Banach spaces. You may want to research this further; see for example Peter Lax's Functional Analysis, Reed and Simon's Methods of Modern Mathematical Physics, Vol. 1 or Extended version, Conway's A Course in Functional Analysis, etc.).

SET 6 - DUE THURSDAY 04/16/2026 BY 5PM

From Chapter 5 (pp 253-255): 1, 9. See definition, examples and hints below.

Definition: A Fourier multiplier operator T on $\mathbb{R}^d$ is a linear operator on $L^2(\mathbb{R}^d)$ determined by a bounded function m (the multiplier) such that T is defined by the formula

$$\widehat{T(f)}(\xi) := m(\xi)\widehat{f}(\xi)$$

for all $\xi \in \mathbb{R}^d$ and any $f \in L^2(\mathbb{R}^d)$.

Examples. The bounded linear operator $P_N : L^2(\mathbb{R}) \rightarrow L^2(\mathbb{R})$ defined by $\widehat{P_N(f)}(\xi) := \chi_{[-N,N]}(\xi)\widehat{f}(\xi)$ is one such operator. In fact is an orthogonal projection.

Another well known one is the *Hilbert Transform* $\mathcal{H} : L^2(\mathbb{R}) \rightarrow L^2(\mathbb{R})$ defined by $\widehat{\mathcal{H}(f)}(\xi) := -isgn(\xi)\widehat{f}(\xi)$. The operator $\mathcal{H}$ is bounded and linear on L^2 . That is bounded foillows from the fact that $-isgn(\xi)$ is bounded point-wise by 1 and Theorem 1.1 that says the Fourier transform is unitary on $L^2(\mathbb{R}^d)$

Hints for 1) Note that a) follows from b) since if a function is in L^2 then it is finite a.e. (in fact you get that $\int |f(x-y)k(y)|dy$ is finite a.e. x .) so prove b) first. Then for c) first prove it for functions in $L^1 \cap L^2$. Then use a density argument: given $f \in L^2$ assume the sequence $f_n \in L^1 \cap L^2$ approximates f in the L^2 sense (L^2 -norm) . Then bound $|\widehat{f * k}(\xi) - \widehat{f_n * k}(\xi)|$ by $\|f_n - f\|_{L^2} \|k\|_{L^1}$ and conclude from here. Finally part d) is the definition of a Fourier multiplication operator applied to part c)

Additional Problem Carefully read and rewrite on your own (justifying and filling the gaps as necessary) Lemma 1.2 on page 209 proving that $\mathcal{S}(\mathbb{R}^d)$ is dense in $L^2(\mathbb{R}^d)$